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Lasers and materials in selective laser sintering

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Research article

Lasers and materials in selective laser sintering

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Keywords

Sintering, Lasers, Simulation, Formed materials

Abstract

Selective laser sintering (SLS) is one of the most rapidly growing rapid prototyping techniques (RPT). This is mainly due to its suitability to process almost any material: polymers, metals, ceramics (including foundry sand) and many types of composites. The material should be supplied as powder that may occasionally contain a sacrificial polymer binder that has to be removed (debinded) afterwards. The interaction between the laser beam and the powder material used in SLS is one of the dominant phenomena that defines the feasibility and quality of any SLS process. This paper surveys the current state of SLS in terms of materials and lasers. It describes investigations carried out experimentally and by numerical simulation in order to get insight into laser-material interaction and to control this interaction properly.

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1. Introduction

Selective laser sintering (SLS) is a rapid prototyping process that allows to generate complex 3D parts by solidifying successive layers of powder material on top of each other (Kruth, 1991). Solidification is obtained by fusing or sintering selected areas of the successive powder layers using thermal energy supplied through a laser beam (Figure 1). A beam deflection system (galvano mirrors or XY table (Van der Schueren and Kruth, 1995a) makes the beam scan each layer according to the corresponding cross section of the part as calculated from a CAD model. A powder deposition system (Van der Schueren and Kruth, 1995b) is used for depositing the successive thin layers of powders (typically 0.1–0.3 mm thickness) in a building container before that layer is laser sintered.

Table I illustrates the recent success of SLS which depicts the largest growth in market share since 1997. This growth is expected to proceed, since – unlike other material additive manufacturing processes mentioned in Table I (Kruth, 1991; Kruth *et al.*, 1998a) – hardly no limitation exists to the materials that might be processed by SLS. The relative importance of SLS will further rise as material additive technology will be used for other applications than the sole production of rapid prototypes and will extend to rapid manufacturing of, e.g. moulds and dies (rapid tooling) or other functional parts in an ever wider range of materials.

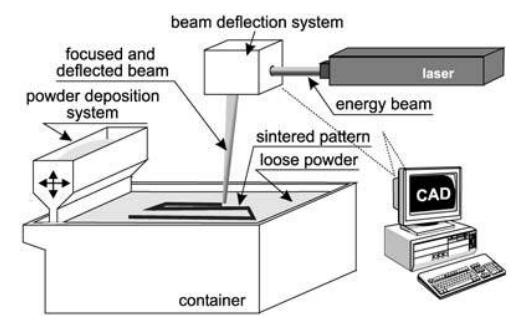
2. Materials in SLS

SLS can be used to process almost any material, provided it is available as powder and that the powder particles tend to fuse or sinter when heat is applied. This is the case for most materials. Powders that depict low fusion or sintering properties can be laser sintered by adding a sacrificial binder material

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Figure 1 Schematic overview of SLS process

(typically a polymer binder) to the basic powder. After sintering the full part, the sacrificial binder can be removed by debinding the “green” part in a thermal furnace. The use of a sacrificial binder allows to enlarge the pallet of laser sinterable materials. However, the range of materials (powders) that can be laser sintered without sacrificial binder is quite large as compared to other rapid prototyping processes.

2.1 SLS of polymers

Polymer powders were the first and are still the most widely applied materials in SLS (Figure 2).

Amorphous polymers, like polycarbonate (PC) powders, are able to produce parts with very good dimensional accuracy, feature resolution and surface finish (depending on the grain size). However, they are only partially consolidated. As a consequence, these parts are only useful for applications that do not require part strength and durability. Typical applications are SLS masters used for manufacturing silicone rubber and cast epoxy moulds (McAlea *et al.*, 1997).

Semi-crystalline polymers, like nylons (polyamide (PA)), on the contrary, can be sintered to fully dense parts with mechanical properties that approximate those of injection moulded parts. On the other hand, the total SLS process shrinkage of these semi-crystalline polymers is typically 3–4 per cent (Grimm, 1997), which

complicates production of accurate parts. The good mechanical properties of these nylon based parts make them particularly suited for high strength functional prototypes. New grades of nylon powders (i.e. Duraform PA12, Schumacher and Levy, 1998) even yield a resolution and surface roughness close to those of PC, making PA also suited for casting silicone rubber and epoxy moulds, even though higher resolutions and smoother surfaces can still be obtained from amorphous powders.

Other polymer-based materials available commercially are acrylic styrene (PMMA/PS) for investment casting and an elastomer for rubber-like applications (Figure 2, top left part).

Table II gives an overview of the mechanical properties of some typical SLS polymer materials supplied by one of the major SLS vendors.

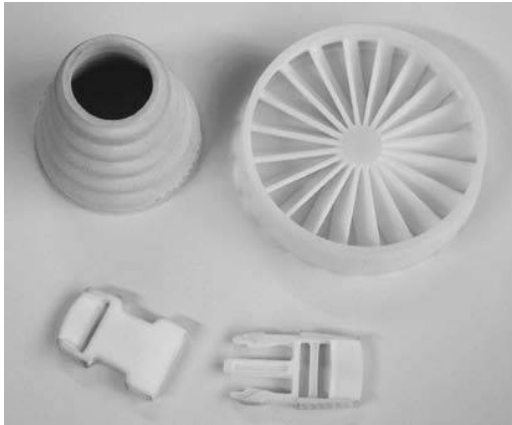
2.2 SLS of reinforced and filled polymers

PA powders can be relatively easily reinforced with other materials in order to further improve their mechanical and thermal properties (Table II). Several grades of glass fibre reinforced PA powders are readily available on the market (Seitz *et al.*, 1997).

PA coated copper powder (Cu-PA) is also available for the production of plastic-metal composite injection tools (Bruning, 1998). This Cu-PA powder mixture contains 70 wt per cent Cu (rest is PA). Compared to plain PA parts (see Table II), Cu-PA SLS parts are 3.5 times heavier (density of 3.45 g/cm³), four times more thermally conductive (1.28 W/m°C) and exhibit a similar tensile strength (34 MPa), but a higher tensile modulus (3.4 GPa). Applications include Cu-PA inserts for injection moulds. Those Cu-PA moulds can be used as laser sintered without the need for removal of the PA phase or without any post-densification process, while still depicting a life time of 200–400 shots.

Table I RP systems unit sales

Process	Sales in 1999	Difference with 1997 (per cent)
Stereolithography (SL)	291 units (24.4 per cent)	+3.5
Fused deposition mod. (FDM)	293 units (24.5 per cent)	+13
Ink jet printing (IJP)	389 units (32.5 per cent)	+47
Laminated object mfg. (LOM)	94 units (9 per cent)	– 42
SLS	115 units (9.6 per cent)	+53

Figure 2 Polymer parts made by SLS

2.3 SLS of metals, hardmetals and cermets

SLS is one of the very few rapid prototyping processes that allow direct manufacturing of metallic components without the use of a polymer binder. Other processes allowing direct production of metallic parts are 3D laser cladding processes (e.g. SDM (Fessler *et al.*, 1998), LENS (Griffith *et al.*, 1996), CMB (Klocke and Clemens, 1996)) and lamination of metal sheets by laser cutting and stacking of sheet material (e.g. LLCC (Dormal *et al.*, 1998), metal sheet lamination (Himmer *et al.*, 1999), CAM-LAM process). Those alternative processes, however, suffer from major limitations in terms of achievable shape complexity and accuracy and are therefore often combined with milling (possibly on a single machine) to remedy those drawbacks. SLS also allows to produce metallic parts using some kind of sacrificial polymer binder, as done with few other RP processes (e.g. SL, 3D printing, LOM). This allows us to further enlarge the range of powders processible by SLS, but requires a furnace post treatment to remove the polymer binder and yield a plain metallic or cermet parts (the so-called debinding). The porosity of laser sintered part may also require a post-densification operation that may be obtained by furnace post-sintering, by pore

infiltration with a metallic or polymeric infiltrant material (Behrendt and Shellabear, 1995; Heymadi and McAlea, 1996), or by hot isostatic pressing (Das *et al.*, 1998; Knight *et al.*, 1996). The following sections will distinguish between those SLS processes that apply polymer binders or infiltrants and those that do not.

2.3.1 SLS of metals and cermets with polymer binder or infiltrant

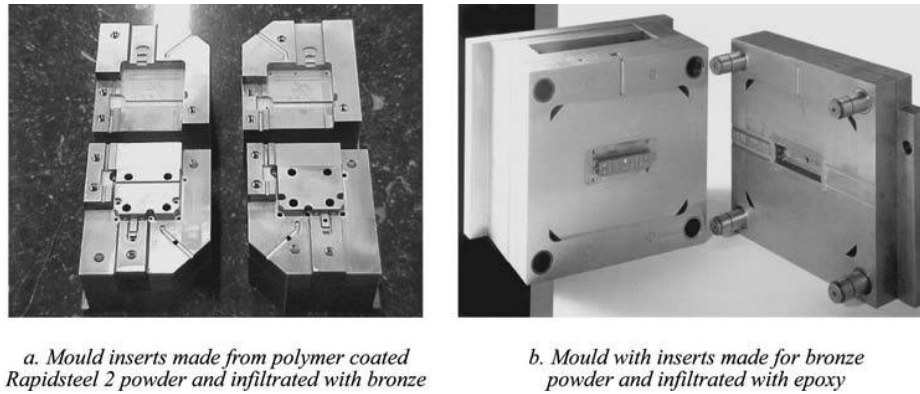
DTM Corporation (Austin, USA) has developed a process that applies polymer-coated steel powders (1080 steel, 316 or 420 stainless steel particles coated/mixed with a thermoplastic/thermoset material) for the SLS of metal parts (Figure 3(a)). During laser sintering, the polymer melts and acts as a binder for the steel particles. After debinding the porous steel part is infiltrated with copper or bronze (McAlea, 2000; McAlea *et al.*, 1997). The resulting material properties are quite close to those of plain steel materials as shown in Table III. Over the years, DTM continuously improved their production process by reducing the number of post-processing cycles and their total duration. For the third generation of RapidSteel powder called LaserForm ST-100, composed of 60 per cent 420 stainless steel and 40 per cent bronze (89Cu-11Sn), debinding and infiltration can be done in a single furnace cycle of about 24 h under pure nitrogen. These developments yield improved material properties of the final SLS parts such as strength, hardness, machinability, weldability, wear rate and thermal conductivity (McAlea, 2000).

Using a similar binder-based SLS process, the University of Texas at Austin produced parts in a SiC-Mg cermet material. It applies SiC particles coated with a proprietary polymer binder to obtain a SiC preform with a typical density of 40 vol per cent (Wohlert and Bourell, 1996). After debinding (at 400°C) the SiC preform becomes quite fragile

Table II Overview of the mechanical properties of some SLS polymer materials (DTM)

	PC	Fine nylon (PA)	Glass filled nylon	Elastomer
E-modulus (MPa)	1200	1,400/1,800 ^a	2,800/4,400 ^a	20
Tensile strength (MPa)	23	36/44 ^a	49/42 ^a	–
Break elongation (per cent)	5	6/22 ^a	1.8 ^a	111
Surface roughness R_a as SLS processed (μm)	7	12/8.5 ^a	15	–

Note: ^aValue for DuraForm PA

Figure 3 Plastic injection moulds or inserts made by SLS**Table III** Overview of the mechanical properties of laser sintered metals (after post-processing and infiltration) and of common plain casted steel

	Reference plain casted steel	RapidSteel 2 powder + bronze infiltration (DTM)	LaserForm ST100 + bronze infiltration (DTM)	Steel/Cu-P/Ni powder (EOS)	Bronze/Ni/P powder + epoxy infiltration (EOS)
E-modulus (GPa)	210	193	153	164	92
Yield strength of 0.2 per cent (MPa)	418	329	326	414	168
Tensile strength (MPa)	500	509	587	491	161

preventing further handling. To improve the strength of the SiC preform, an additional firing step (1,100°C, 2 h, formation of SiO₂ layer) was utilized followed by (pressureless) infiltration (670°C) with a Mg-based (AZ91D) die casting alloy (Wohlert and Bourell, 1996).

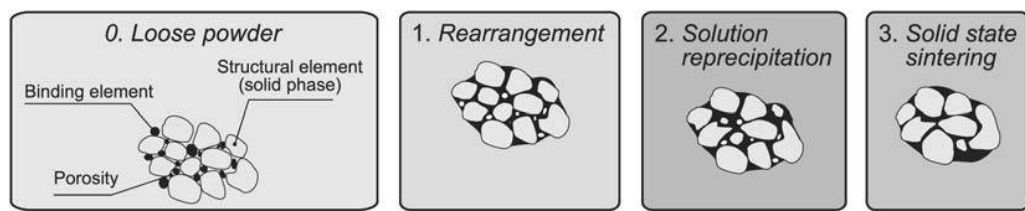
EOS GmbH (Munich, Germany) avoids the use of a polymer binder by directly sintering metal powders with a low melting point, i.e. bronze-nickel based powders (EOS-Cu 3201 containing Cu-Sn, Cu-P and Ni particles) developed by Electrolux Co. (Behrendt and Shellabear, 1995). An example is given in Figure 3(b). After SLS, the part is infiltrated with epoxy resin to fill in the porosities. Hence the final part is a bronze-epoxy composite, rather than a plain metallic part and its mechanical and thermal properties are limited (Table III). Infiltration with a metal like Cu or bronze is not possible in this case, since the green part would melt during infiltration. Recently, EOS put into market a new powder (EOS-DMLS Steel 50-V1 containing steel, Cu-P and Ni particles) yielding improved mechanical properties (PM update, 1998). The SLS part is about 70 per cent dense and thus can be used as such for inserts and small mould components.

2.3.2 SLS of metals and hardmetals/cermets by liquid phase sintering

Many research institutes study the possibility of directly laser sintering metal and ceramic powders without the use of any polymer component (Bourell *et al.*, 1992; Coremans *et al.*, 1996; Das *et al.*, 1996; Kruth *et al.*, 1996, 1997; Laoui *et al.*, 1998; O'Neill *et al.*, 1999; Song and König *et al.*, 1997; Van der Schueren and Kruth, 1994). For this purpose, several approaches for binding powder particles together using laser beam energy have been investigated at the University of Leuven including: solid state vs liquid phase sintering (LPS), loose vs pre-coated metal binder phase, mixed vs milled powders (Kruth *et al.*, 1996; Laoui *et al.*, 1998).

The basic material used in LPS consists of a mixture of two metal powders (or of a metal and a ceramic powder in case of hardmetals and cermets), i.e. a high melting point metal or ceramic, called the structural material, and a low melting point metal, called the binder. Applying heat to the system causes the binder to melt and flow into the pores formed by the non-molten particles. The classical stages of LPS are schematically shown in Figure 4.

Besides the very good mechanical properties generally obtained with two-phase

Figure 4 Different stages of LPS

or composite LPS materials, the main advantage of LPS is the very fast initial binding occurring during laser heating. This binding is based on capillary forces, which can be very high: the reaction speed in this stage is determined by the kinetics of the solid-melt transformation. This transformation is several orders of magnitudes faster than solid-state diffusion of atoms occurring in the solid phase sintering.

Once the binder metal is molten and spread out into the solid lattice, the system cools down as the laser beam moves away and the situation is frozen. Only the first stage (rearrangement) of the LPS mechanism takes place during laser sintering because of the very short laser-material interaction time (fraction of a second).

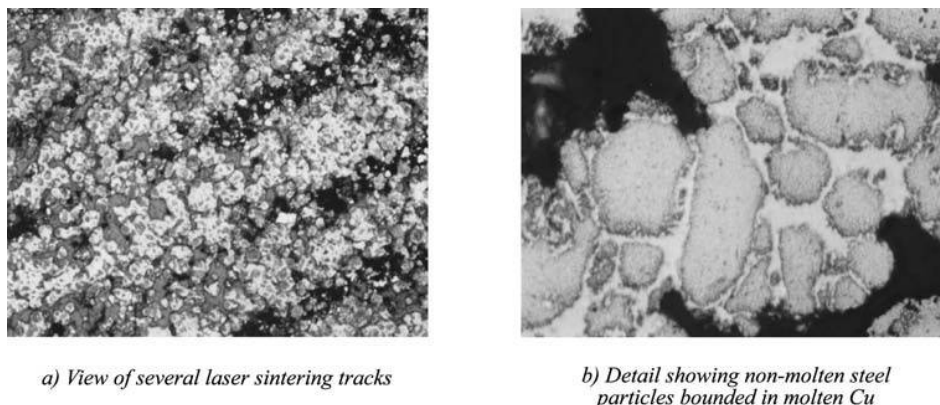
Early studies to sinter steel or iron powder, mixed up with copper grains serving as binder material, were performed at several universities (Coremans *et al.*, 1996; Miani *et al.*, 2000; Song and Konig, 1997; Van der Schueren and Kruth, 1994). Figure 5 shows an example of a steel-copper powder mixture laser sintered at the University of Leuven. Such LPS green part has enough strength to withstand a post-processing cycle to bring the part to full density. This post-processing consists of a furnace post-sintering to proceed with the next

stages of LPS shown schematically in Figure 4 (solution reprecipitation and solid state sintering) or an infiltration with a low melting point metal (typically copper or bronze).

The universities of Texas at Austin (USA) (Jepson *et al.*, 1999) and Leuven (Belgium) (Kruth *et al.*, 1997; Laoui *et al.*, 1998) and some German Fraunhofer research centres (Private communication, 1999) succeeded to laser sinter hardmetals (i.e. cemented carbides) and cermets by SLS. Figure 6(a) shows a typical microstructure of a WC-9 wt per cent Co powder mixture sintered by a CO₂ laser showing a good bounding between the WC particles surrounded by a Co-binder and the presence of large pores (50–60 per cent). After infiltration, Cu filled these open porosities (Figure 6(b)). Examples of WC-Co parts sintered at the University of Leuven are depicted in Figure 7.

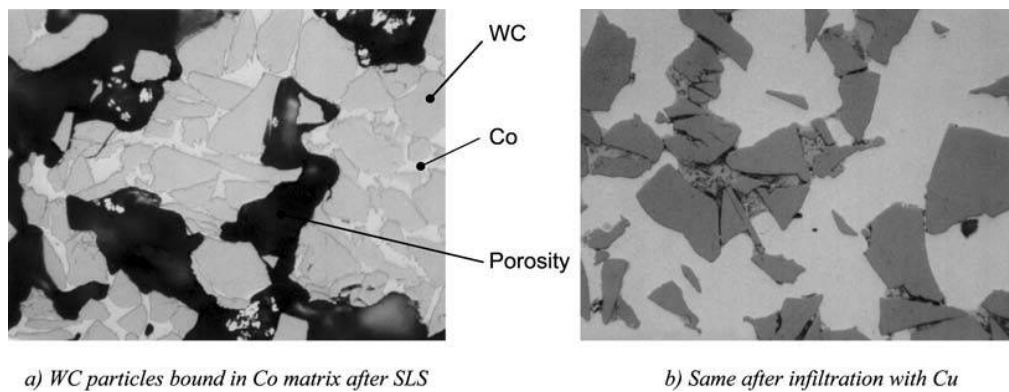
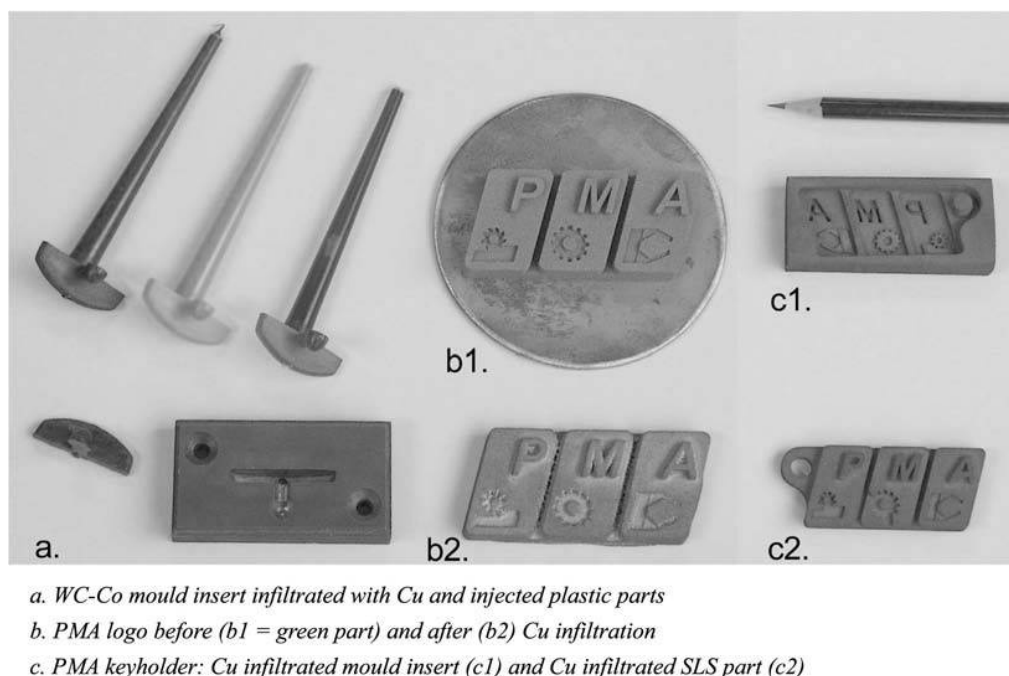
A wide variety of powder combinations have been investigated in Leuven using LPS mechanism, including: Fe-Cu, Cu-coated Fe, Fe₃C-Fe, stainless steel-Cu, WC-Co, co-coated WC, WC-Cu, WC-CuFeCo, TiC-Ni/Co/Mo, TiB₂-Ni, ZrB₂-Cu, etc. (Laoui *et al.*, 1999b).

Significant improvements were obtained when using composite powders, obtained by mechanical alloying a mixture of two powder

Figure 5 SLS of steel-copper powder mixture

a) View of several laser sintering tracks

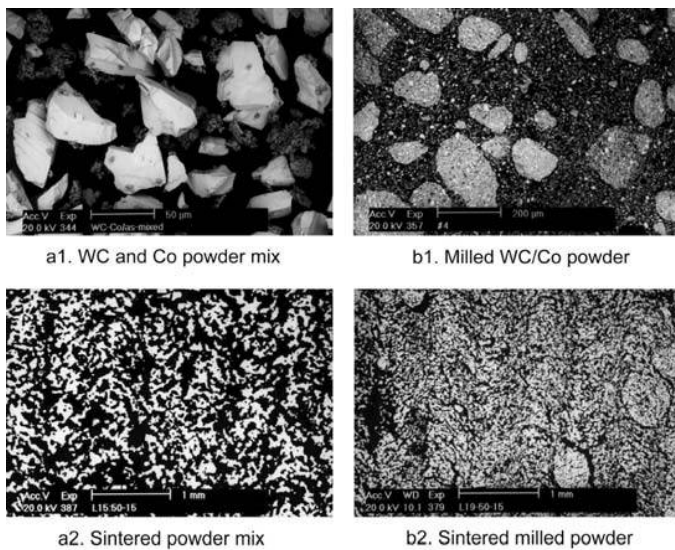
b) Detail showing non-molten steel particles bounded in molten Cu

Figure 6 SLS of WC-9 wt per cent Co powder mixture, followed by Cu infiltration**Figure 7** Various WC-Co parts made by SLS

materials (e.g. mixture of WC and Co powder particles as shown in Figure 8(a1)). During mechanical alloying, the individual powder particles are milled, repeatedly fractured and welded together. This results in new powder particles depicting a fine micrograin composite structure in which two phases (i.e. the two original materials, here WC and Co) can still be identified (Figure 8(b1)). Laser sintering WC-9 wt per cent Co mechanically alloyed powder (Figure 8(b2)) resulted in higher green densities with better surface roughness as compared to direct laser sintering of a WC-9 wt per cent Co powder mixture not subjected to mechanical alloying (Figure 8(a2)) (Laoui *et al.*, 1999a).

2.3.3 SLS of metals through melting

The Fraunhofer Institute ILT (Aachen, Germany) applied a 300 W Nd:YAG laser to completely melt metal powders (bronze, steel, stainless steel such as 316L) deposited in a standard way using a wiper (scraper) and producing directly dense parts (density > 95 per cent) (Klocke *et al.*, 1996). Due to the tendency of molten metal to form droplets and minimise surface energy, careful control of the process parameters is needed. Moreover, overhangs with angles higher than 60° could not be built with this process. When this process was used to sinter Al-30 per cent Si, a maximum density of 90–95 per cent was obtained (Private communication, 1999).

Figure 8 SLS of mixed (unalloyed) and milled (alloyed) WC-Co powder

Note: White = WC; Grey = Co, Black = pores

EOS recently came to the market with a plain steel powder that is laser sintered through melting. The average particle size is $50\text{ }\mu\text{m}$, but an enhanced steel powder with $20\text{ }\mu\text{m}$ size is announced.

Osaka University (Japan) utilised a pulsed Nd:YAG laser (50 W mean power, 3 kW maximum peak power) to melt pure Ti spherical powders (200 and $50\text{ }\mu\text{m}$ average particle size) to produce medical parts (dental crowns and bone models) (Abe *et al.*, 2000). For the coarse Ti powder, the SLS part delivered a maximum relative density of 84 per cent yielding a maximum tensile strength of 70 MPa. Using fine Ti powder ($25\text{ }\mu\text{m}$), a higher relative density (maximum 93 per cent) was achieved with a tensile strength of 150 MPa. Due to the presence of remaining porosity, the tensile strength of these SLS parts is still lower than that of bulk pure Ti material (275–481 MPa) (Abe *et al.*, 2000).

2.4 SLS of ceramics

The Fraunhofer Institute IPT used the SLS process in an attempt to produce directly ceramic parts without polymer binder material. The absence of any binder element makes the ceramic laser sintered part very fragile and viable to breakage. Due to the short reaction time involved in SLS, solid state sintering is not feasible. To sinter SiC powder material, a sufficient amount of laser energy was supplied to induce high local

temperatures leading to a partial disintegration of SiC particles into Si and C. The free Si then oxidises and forms SiO_2 , which plays a role of a binder between the SiC particles (Klocke and Wirtz, 1997; Klocke *et al.*, 1996). After laser sintering, the SiC parts could be infiltrated with Si and reaction bonded to full density. Zirconium silicates were also laser sintered by almost fully melting the powder particles forming large agglomerates (Klocke and Wirtz, 1997, 1998). Similar to DTM's polymer coated powder process, graphite coated with phenolic resin was also processed by SLS by melting only the polymer binder, which is burned out afterwards. However, the resulting graphite part becomes very fragile (Klocke and Wirtz, 1997).

2.5 SLS of foundry sand

The two commercial SLS machine vendors (DTM and EOS) offer sand powders that can be laser sintered in order to produce foundry sand moulds. DTM, for instance, offers both Zr and Si sand: SandForm ZrII and Si (Seitz, 1998). Key characteristics include Shell Foundry Sand of given AFS grain fineness number ($\text{GFN}\# = 97$ for Si and 99 for ZrII) and dimensional tolerances of 0.5 mm. SandForm Si, used predominantly for Al castings, is based on silica, which is prevalent in the market and has a low density. SandForm ZrII can be used for both Al and Fe castings and its binder system matches silica.

3. Lasers and materials

Different kind of lasers are applied in SLS. Commercial SLS machines (DTM and EOS) are all equipped with CO_2 lasers with maximum power ratings between 50 and 200 W. The University of Leuven developed two SLS machines equipped, respectively, with continuous wave Nd:YAG lasers of 300 and 500 W (Van der Schueren and Kruth, 1995a). So did others (Klocke and Wirtz, 1996; Laoui *et al.*, 1999b; Song and Konig, 1997). The University of Liverpool explored the use of Q-switched Nd:YAG and short-pulse Cu-vapour lasers (O'Neill *et al.*, 1999). The University of Connecticut and University of Manchester used a 60 W diode laser of 810 nm (Li *et al.*, 1998; Manzur *et al.*, 1996). In future, other type of lasers might

show up like diode pumped solid-state lasers and others.

The choice of a proper SLS laser might not be independent of the material to be sintered. Varying SLS process parameters like laser wavelength (i.e. type of laser), laser energy (i.e. power, scan speed and scan spacing) and powder characteristics (particle size, powder composition, mixing, etc.), greatly influences the resulting part properties such as surface quality or part density (Laoui *et al.*, 1998, 1999a). Optimally, the laser wavelength should be adapted to the powder material to be sintered, since laser absorption greatly changes with the material and frequency or wavelength of the laser light (Olsen and Femming, 1989; Tolochko *et al.*, 2000). Figure 9 shows how absorption coefficients varies versus wavelength for a solid polymer (PC) and metal (Cu). Laser absorption in powders is usually larger than in solid material. This is due to multiple reflection and absorption of the laser beam trapped in the pores of the powder (Figure 12). In order to distinguish between the basic material absorption coefficient of solid material upon a single impingement of a laser ray and the total absorption of a powder upon multiple impingement (i.e. multiple partial absorption and partial reflection), let us call the former as “material absorption” and the later as “powder incoupling”. Results of measurements of laser incoupling in powders performed with an integrating sphere are reported in Table IV (Tolochko *et al.*, 2000). Measurements with single component powders (Ni-alloy) of different particle sizes proved that the particle size has little influence

on the incoupling. This statement is confirmed by simulations (see further), but is no longer true when mixing two powders of different material and different grain size (see simulation results below).

CO₂ lasers with wavelength of 10.6 μm are well suited for sintering polymer powders, as polymers depict high absorption at far infrared or long wavelength (see Table IV and Figure 9). The same is true for oxide ceramics, but no longer for carbide ceramics that better absorb at Nd:YAG wavelength of 1.06 μm (Table IV). It is also well known that metals absorb much better at short wavelength (Table IV and Figure 9). This is why Nd:YAG lasers may outperform CO₂ lasers for metallic materials. The latter is particularly true for LPS of metallic materials and has been confirmed by studies of the University of Leuven in which Fe-Cu metal powders and WC-Co hardmetal or cermet powders were alternatively sintered with both laser types (Kruth *et al.*, 1998b). Experiments proved that for the same amount of energy (or at similar settings for laser power and scan speed), a Nd:YAG laser results in higher green part density (Figure 10), a larger sintering depth (allowing thicker layers to be sintered, hence reducing production time) and a higher energetic process efficiency. Figure 11 also shows that, for those materials, the processing window where a YAG laser yields proper LPS is larger than that of a CO₂ laser, as well as in terms of allowable range of supplied energy as of powder composition. This allows a wide variation of the processing parameters (power and scan speed) or powder composition (here mixture ratio between Fe and Cu), while still ensuring good sintering results (i.e. proper LPS, yielding fine composite structure and better part properties as compared to true melting). The large processing window enforces good process controllability and reliability.

The small LPS processing window in the case of CO₂ is explained by the small difference in melting points between the structural element (stainless steel) and the binder element (Cu) and by the high reflectivity of Cu particles for CO₂ laser light. This induced mostly simultaneous melting of both elements, rather than preferential melting of the Cu binder and resulted in reduced mechanical properties. Owing to the larger difference in their melting points, such behaviour was not observed with WC-Co

Figure 9 Absorption of plain solid materials (PC and copper) versus wavelength

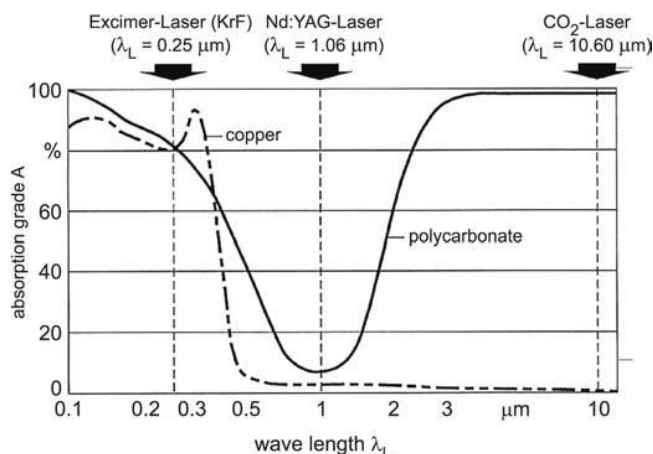


Table IV Incoupling (absorption) of powder materials for Nd:YAG and CO₂ wavelength

Powder materials	Nd:YAG (1.06 μm)	CO ₂ (10.6 μm)
Metals		
Cu (absorption in solid Cu)	59 per cent (2-10 per cent*)	26 per cent (1 per cent)
Fe (absorption in solid Fe)	64 per cent (30 per cent)	45 per cent (4 per cent)
Sn	66 per cent	23 per cent
Ti	77 per cent	59 per cent
Pb	79 per cent	—
Co-alloy (1 per cent C; 28 per cent Cr; 4 per cent W)	58 per cent	25 per cent
Cu-alloy (10 per cent Al)	63 per cent	32 per cent
Ni-alloy 1 (13 per cent Cr; 3 per cent B; 4 per cent Si; 0.6 per cent C)	64 per cent	42 per cent
Ni-alloy 2 (15 per cent Cr; 3.1 per cent Si; 0.8 per cent C)	72 per cent	51 per cent
Ceramics		
ZnO	2 per cent	94 per cent
Al ₂ O ₃	3 per cent	96 per cent
SiO ₂	4 per cent	96 per cent
SnO	5 per cent	95 per cent
CuO	11 per cent	76 per cent
SiC	78 per cent	66 per cent
Cr ₃ C ₂	81 per cent	70 per cent
TiC	82 per cent	46 per cent
WC	82 per cent	48 per cent
Polymers		
Polytetrafluoroethylene	5 per cent	73 per cent
Polymethylacrylate	6 per cent	75 per cent
Epoxyether-based polymer	9 per cent	94 per cent
Mixtures		
Fe-alloys (3 per cent C; 3 per cent Cr; 12 per cent V) + 10 per cent TiC	65 per cent	39 per cent
Fe-alloys (1 per cent C; 14 per cent Cr; 10 per cent Mn; 6 per cent Ti) + 66 per cent TiC	79 per cent	44 per cent
Ni-alloy 2 (95 per cent) + Epoxyether-based polymer (5 per cent)	68 per cent	54 per cent
Ni-alloy 2 (25 per cent) + Epoxyether-based polymer (75 per cent)	23 per cent	76 per cent

Note: *Value varying with gloss, surface roughness, surface oxidation, etc.

Source: Tolochko *et al.* (2000)

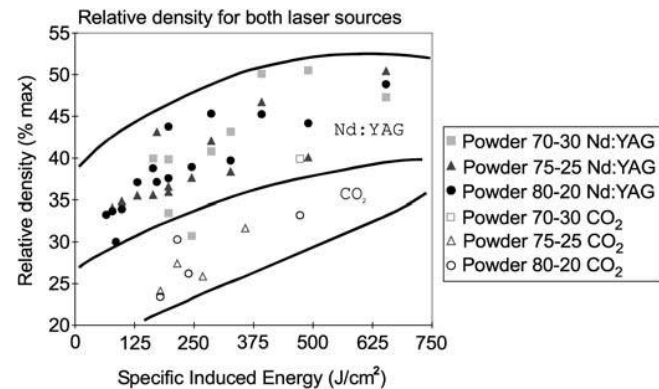
powder mixtures where most of the SLS tests resulted in LPS without much difficulty. Further details about these results can be found elsewhere (Kruth *et al.*, 1998b; Laoui *et al.*, 1998, 2000a).

4. Modelling and simulation of material-laser behaviour

Several simulation models have been developed to get a better understanding of

the SLS process and of the interaction between the laser beam and powder material. These models not only provide a theoretical explanation to the phenomena discussed above, but also have a major practical significance. Indeed, the final properties of parts fabricated by SLS very much depend on the process parameters as well as on the powder characteristics. Simulation allows to identify the processing window and to select proper SLS parameters without need for extensive testing for each powder.

Figure 10 Relative density of stainless steel-Cu SLS parts processed by CO₂ and Nd:YAG lasers



Many researchers apply thermal finite element methods to model the SLS process (Childs *et al.*, 1999, 2000; Nelson, 1993; Wang and Kruth, 2001; Weissman and Hsu, 1991). Those models need a heat source to be defined, which in turn requires to postulate the powder absorption (incoupling) and the depth at which the absorption takes places. FEM models also need to assume the heat conductivity in the powder, which is generally very low and even negligible (Wang and Kruth, 2001). In most FEM models, a surface heat source is assumed, neglecting the penetration of the laser beam into the powder bed (porosity). A volumetric heat source has been applied by Childs *et al.* (2000) and Wang and Kruth (2001), demonstrating that the depth of the heat source into the powder bed largely influences the quality of the simulation.

A ray tracing (RC) model has been developed at the University of Leuven (Laoui *et al.*, 2000b; Wang and Kruth, 2000). It is based on a geometrical simulation of the impingement, reflection and absorption of a large number of rays (representing the laser beam) onto/into a powder bed (Figure 12).

Each time a ray hits a powder particle, the amount of absorbed and reflected energy is calculated and the reflected ray is traced further along its way into the pores of the powder bed. This model allows to calculate:

- the powder incoupling E (i.e. total absorption) versus the material absorption coefficient α : see Figure 13 (relation between incoupling and material absorption coefficient for single component powder) and Table V (total incoupling into Fe-Cu and WC-Co powder mixture for Nd:YAG and CO₂ laser);
- the amount of energy absorbed by each powder material in the case of powder mixtures Table V;
- the laser beam penetration and the absorption profile versus the depth into the powder bed (Figure 14);
- the sintering depth and width of a single laser track (Figure 15).

The RC simulations demonstrate the following.

- Incoupling in powders is much higher than in solid material, mainly for materials with low absorption coefficients where the increase is more than an order of magnitude (see Figure 13 and experimental data on Cu in Table IV).
- The maximum of the absorption profile of a CO₂ laser lies deeper under the powder bed surface than for a Nd:YAG laser (Figure 14). However, the larger total incoupling with Nd:YAG laser (see Table V) still results in a larger sintering depth with that laser.
- In the case of powder mixtures, the relative particle size of the two powder components largely influences the relative amount of energy absorbed by each powder material. This is illustrated in

Figure 11 Processing window for stainless steel-Cu sintered by CO₂ and Nd:YAG laser

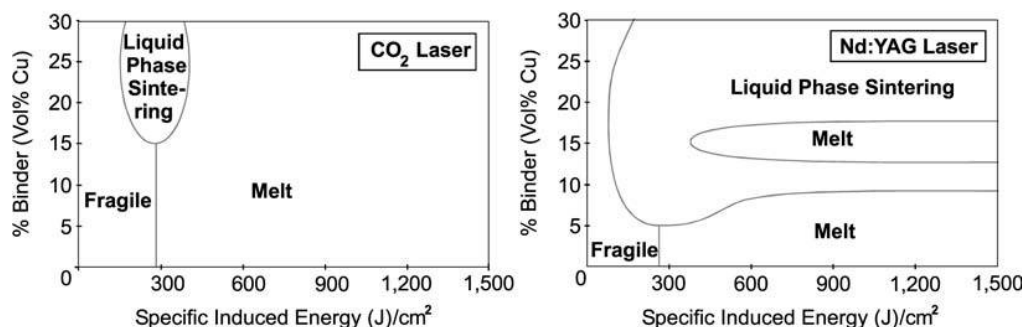


Figure 12 2D illustration of the RC model: simulation over a depth of 1 mm and width of 1 mm of powder bed

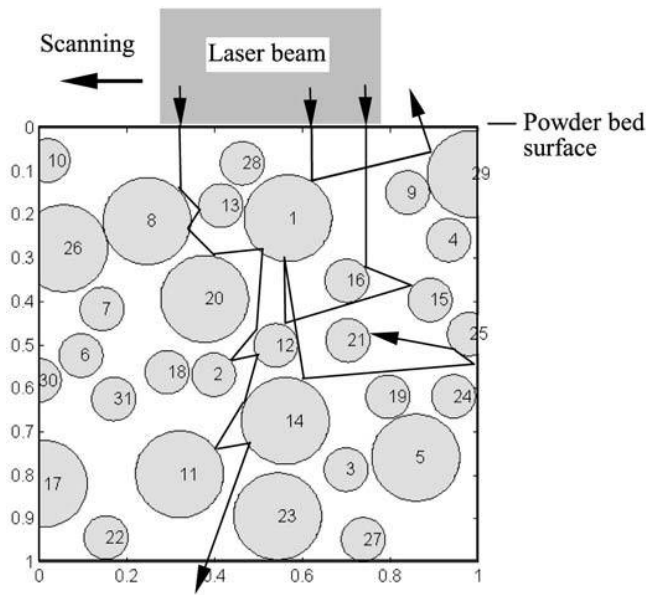


Table VI for a mixture of Fe and Cu particles regularly stacked according to a phase centred structure and sintered with a Nd:YAG laser (Van der Schueren and Kruth, 1994). When mixing Fe and Cu particles of equal size or diameter (Simulation 1), the less reflective Fe will absorb more energy and achieve its melting point before the high reflective Cu starts melting (comparing required and absorbed energies), even though Fe has a much higher melting point than Cu (1,500°C versus 1,083°C). To enforce proper LPS,

where only the Cu melts and binds the solid Fe particles, the Cu particles should be taken smaller than the Fe particles (Simulations 2, 3 and 4).

- The RC model demonstrates a good accuracy when comparing the results to the experimental data (Figure 13, compare also to Cu and Fe values in Table IV). Comparison of sintering depth and width calculated solely with the RC model and those calculated with a combined RC-FEM model (RC model provides heat input profile for thermal FEM simulation) show little difference (Wang and Kruth, 2001) (Figure 15). This indicates that, in most cases, FEM simulations do not add much to the precision of the RC model. This is due to the almost negligible heat conductivity in porous powder beds.

As shown in Table VI, the RC model further allows to analyse the influence of various parameters like particle diameter, mixture ratio, type of material (i.e. material absorption coefficient), etc.

5. Conclusion

SLS holds an important place among the wide variety of rapid prototyping processes (Kruth, 1991; Kruth *et al.*, 1998a). One of its major advantages is the ability to process about any (powder) material: polymers (elastomers, amorphous and semi-crystalline technical polymers), metals, hardmetals/cermets,

Figure 13 Total energy incoupling versus the material absorption coefficient: modelling and experiments

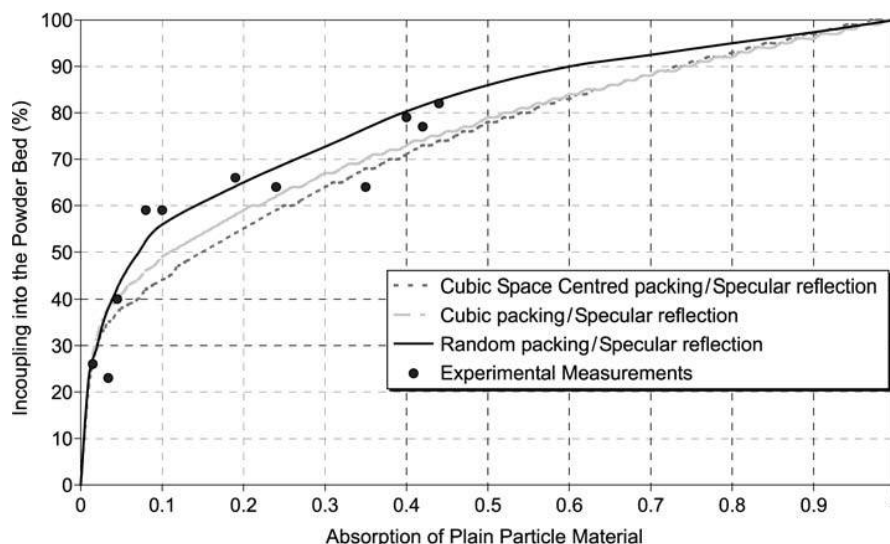


Table V Incoupling E (per cent) when sintering powder mixtures with porosity of about 75 per cent of full density

Powder (grain size)	Type of laser	$\lambda(\mu\text{m})$	$\alpha_{\text{Fe, resp. WC}}$ (per cent)	$\alpha_{\text{Cu, resp. Co}}$ (per cent)	E_{total} (per cent)	E_{Fe} (per cent)	E_{Cu} (per cent)
Fe-30 wt per cent Cu (Fe = 50 μm ; Cu = 30 μm)	Nd:YAG	1.06	30	10	66	54	12
	CO ₂	10.6	3.5	1.5	26.5	21	6
WC-9 wt per cent Co (WC = 50 μm ; Co = 20 μm)	Nd:YAG	1.06	55	31	80	61	19
	CO ₂	10.6	??	2.6	–	–	–

Figure 14 Energy absorption profiles versus the depth into the powder bed

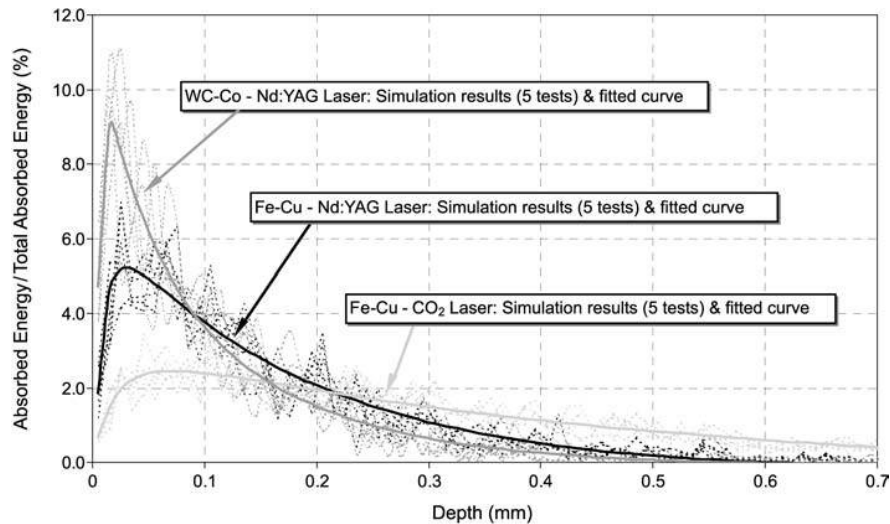
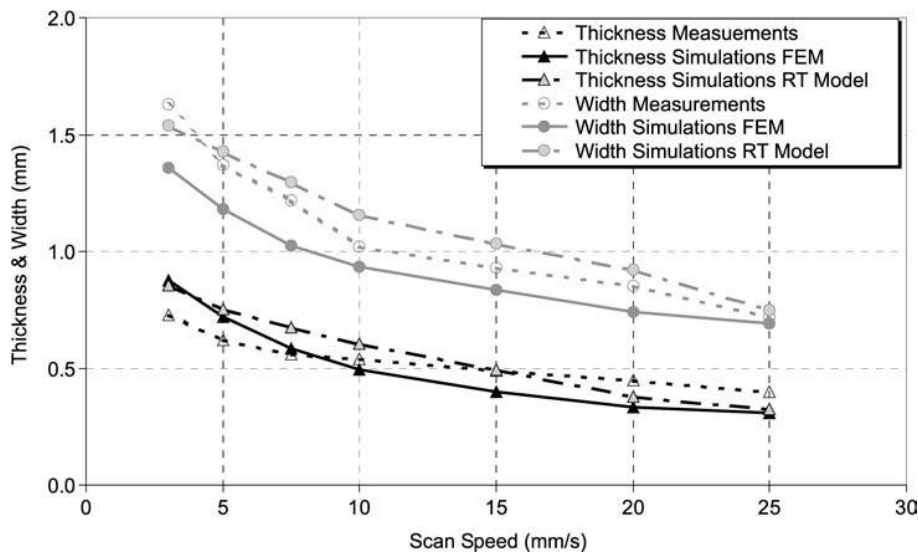


Figure 15 Track thickness and width when sintering Fe-Cu powder with Nd:YAG laser



ceramics and sand. Among others, SLS is well suited to produce a variety of composite materials: glass reinforced polymers, metal/polymer composite (e.g. Cu/PA), metal/metal composites (e.g. Fe/Cu), cermets (e.g. WC-Co) and others. In many cases, standard off-the-shelf powder materials can be used, without the need to develop

dedicated powders. Quite some developments, however, aim to develop powders specially tuned for selecting laser sintering. The purpose is to have powders depicting finer particle size (thinner layers, better resolution and finer roughness), better flowability (for easier powder layer deposition); less shrinkage during laser

Table VI Simulation results for mixtures Fe and Cu powder particles regular stacked as successive layers of grains (phase centred packing). A single sintered powder layer contains several layers of grains

Simulation parameters				Simulation results	
Parameter	Fe	Cu	Energy mJ (per cent)	Fe	Cu
1 Diameter (μm)	50	50	Required for melting	0.45	0.32
Absorptivity (per cent)	30	10	Absorbed by layer 1	0.45 (25.3 per cent)	0.037 (2.1 per cent)
Mixture ratio (per cent)	80	20	Absorbed by layer 2	0.38 (21.4 per cent)	0.032 (1.8 per cent)
2 Diameter (μm)	50	10	Required for melting	0.45	0.0026
Absorptivity (per cent)	30	10	Absorbed by layer 1	0.031 (25.3 per cent)	0.0026 (2.1 per cent)
Mixture ratio (per cent)	80	20	Absorbed by layer 2	0.026 (21.4 per cent)	0.0022 (1.8 per cent)
3 Diameter (μm)	50	30	Required for melting	0.45	0.069
Absorptivity (per cent)	30	10	Absorbed by layer 1	0.31 (20 per cent)	0.069 (4.4 per cent)
Mixture ratio (per cent)	60	40	Absorbed by layer 2	0.26 (16.5 per cent)	0.058 (3.7 per cent)
4 Diameter (μm)	50	30	Required for melting	0.45	0.069
Absorptivity (per cent)	30	20	Absorbed by layer 1	0.41 (24.5 per cent)	0.069 (4.1 per cent)
Mixture ratio (per cent)	80	20	Absorbed by layer 2	0.32 (19.4 per cent)	0.054 (3.2 per cent)

Source: Van der Schueren and Kruth (1994)

sintering (no part warpage), higher green density, higher strength, etc.

The quality of laser sintered parts, however, greatly depends on proper selection of the processing parameters. The latter not only comprises the adjustable machine parameters (like laser power, spot size, scanning speed, etc.), but also the type of laser (wavelength) and the powder composition (materials, mixture ratios, grain sizes, etc.). This is because the SLS process is not only controlled by the amount and speed of energy supply, but to a great extent by the basic laser-material interaction. Experimental research is going on to better understand this interaction. Simulation models proved beneficial for demonstrating the influence of the various parameters. Those models are the essential tools for identifying proper parameters without extensive testing.

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